

```
In [18]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [19]: df = pd.read_csv('heart.csv' )
```

```
In [20]: df.head()
```

```
Out[20]:
```

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	Exer
--	-----	-----	---------------	-----------	-------------	-----------	------------	-------	------

0	40	M	ATA	140	289	0	Normal	172	
1	49	F	NAP	160	180	0	Normal	156	
2	37	M	ATA	130	283	0	ST	98	
3	48	F	ASY	138	214	0	Normal	108	
4	54	M	NAP	150	195	0	Normal	122	



```
In [21]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 918 entries, 0 to 917
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                    918 non-null   int64
1   Sex                    918 non-null   object
2   ChestPainType          918 non-null   object
3   RestingBP              918 non-null   int64
4   Cholesterol             918 non-null   int64
5   FastingBS              918 non-null   int64
6   RestingECG             918 non-null   object
7   MaxHR                  918 non-null   int64
8   ExerciseAngina         918 non-null   object
9   Oldpeak                918 non-null   float64
10  ST_Slope               918 non-null   object
11  HeartDisease           918 non-null   int64
dtypes: float64(1), int64(6), object(5)
memory usage: 86.2+ KB
```

```
In [22]: df.describe()
```

Out[22]:

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	Oldpeak	HeartDisea
count	918.000000	918.000000	918.000000	918.000000	918.000000	918.000000	918.0000
mean	53.510893	132.396514	198.799564	0.233115	136.809368	0.887364	0.5533
std	9.432617	18.514154	109.384145	0.423046	25.460334	1.066570	0.4974
min	28.000000	0.000000	0.000000	0.000000	60.000000	-2.600000	0.0000
25%	47.000000	120.000000	173.250000	0.000000	120.000000	0.000000	0.0000
50%	54.000000	130.000000	223.000000	0.000000	138.000000	0.600000	1.0000
75%	60.000000	140.000000	267.000000	0.000000	156.000000	1.500000	1.0000
max	77.000000	200.000000	603.000000	1.000000	202.000000	6.200000	1.0000

In [23]: `df.isnull().sum()`

Out[23]:

Age	0
Sex	0
ChestPainType	0
RestingBP	0
Cholesterol	0
FastingBS	0
RestingECG	0
MaxHR	0
ExerciseAngina	0
Oldpeak	0
ST_Slope	0
HeartDisease	0

dtype: int64

In [24]: `df.dtypes`

Out[24]:

Age	int64
Sex	object
ChestPainType	object
RestingBP	int64
Cholesterol	int64
FastingBS	int64
RestingECG	object
MaxHR	int64
ExerciseAngina	object
Oldpeak	float64
ST_Slope	object
HeartDisease	int64

dtype: object

In [25]: `df.duplicated().sum()`Out[25]: `np.int64(0)`

```
In [26]: df.drop(columns=['Oldpeak' , 'ST_Slope'] , axis= 1, inplace=True)
```

```
In [27]: df.head()
```

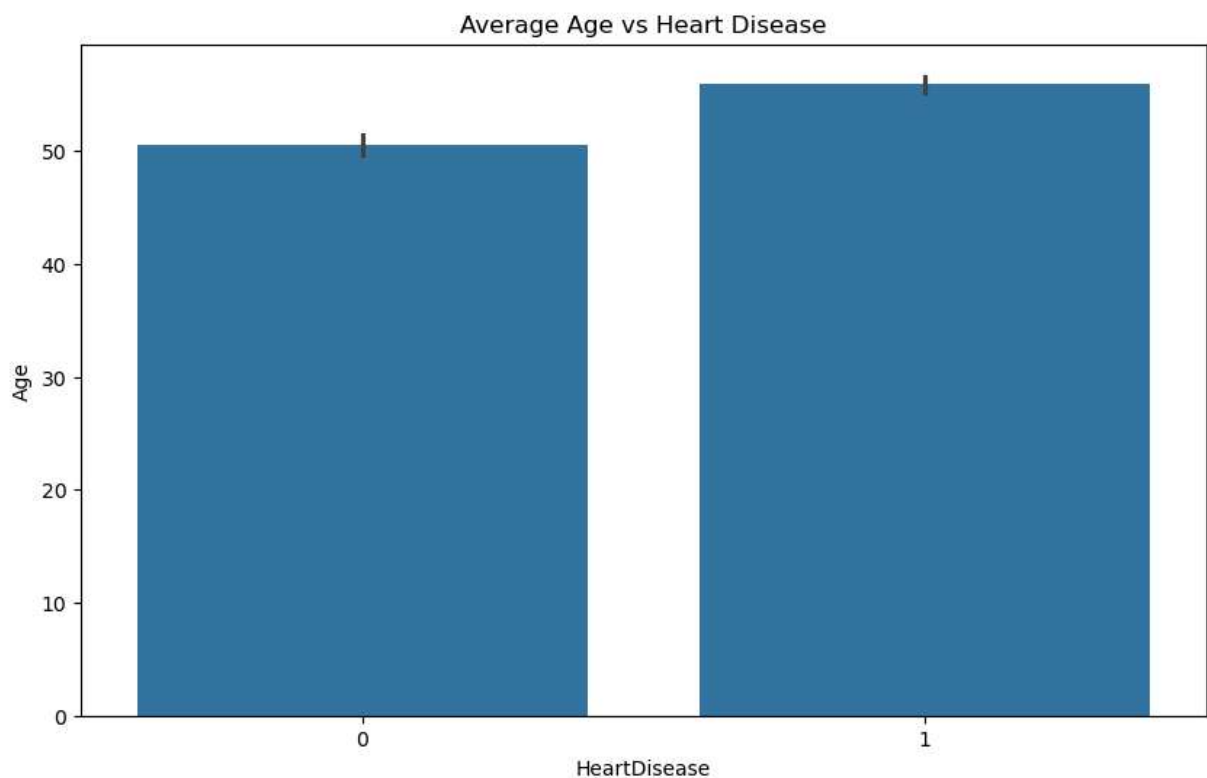
```
Out[27]:
```

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina
0	40	M	ATA	140	289	0	Normal	172	0
1	49	F	NAP	160	180	0	Normal	156	0
2	37	M	ATA	130	283	0	ST	98	0
3	48	F	ASY	138	214	0	Normal	108	0
4	54	M	NAP	150	195	0	Normal	122	0

```
In [28]: df['HeartDisease'].value_counts()
```

```
Out[28]: HeartDisease
1      508
0      410
Name: count, dtype: int64
```

```
In [29]: plt.figure(figsize = (10 , 6))
sns.barplot(x='HeartDisease', y='Age', data=df)
plt.title("Average Age vs Heart Disease")
plt.show()
```



```
In [30]: df.columns
```

```
Out[30]: Index(['Age', 'Sex', 'ChestPainType', 'RestingBP', 'Cholesterol', 'FastingBS',
              'RestingECG', 'MaxHR', 'ExerciseAngina', 'HeartDisease'],
              dtype='object')
```

```
In [31]: df.describe()
```

```
Out[31]:
```

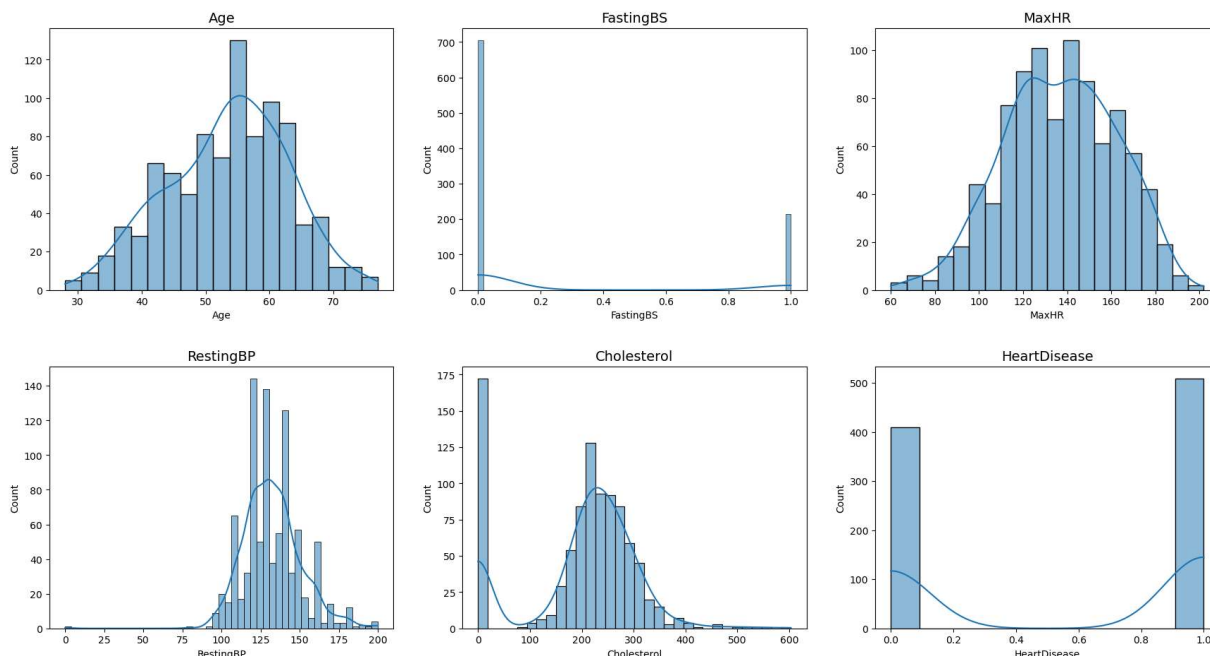
	Age	RestingBP	Cholesterol	FastingBS	MaxHR	HeartDisease
count	918.000000	918.000000	918.000000	918.000000	918.000000	918.000000
mean	53.510893	132.396514	198.799564	0.233115	136.809368	0.553377
std	9.432617	18.514154	109.384145	0.423046	25.460334	0.497414
min	28.000000	0.000000	0.000000	0.000000	60.000000	0.000000
25%	47.000000	120.000000	173.250000	0.000000	120.000000	0.000000
50%	54.000000	130.000000	223.000000	0.000000	138.000000	1.000000
75%	60.000000	140.000000	267.000000	0.000000	156.000000	1.000000
max	77.000000	200.000000	603.000000	1.000000	202.000000	1.000000

```
In [32]: plt.figure(figsize=(18, 10))

def plotting(var, num):
    plt.subplot(2, 3, num)
    sns.histplot(df[var], kde=True, bins='auto')
    plt.title(var, fontsize=14)

plotting('Age', 1)
plotting('FastingBS', 2)
plotting('MaxHR', 3)
plotting('RestingBP', 4)
plotting('Cholesterol', 5)
plotting('HeartDisease', 6)

plt.tight_layout(pad=3)
plt.show()
```



```
In [33]: #changing thr value of restingBP and Cholesterol
BP_mean = df.loc[df['RestingBP'] != 0, 'RestingBP'].mean()
print(BP_mean)
```

132.54089422028352

```
In [34]: df['RestingBP'] = df['RestingBP'].replace(0 , BP_mean)
df['RestingBP'] = df['RestingBP'].round(2)
```

```
In [35]: ch_mean = df.loc[df['Cholesterol'] != 0 , 'Cholesterol'].mean()
print(ch_mean)
```

244.6353887399464

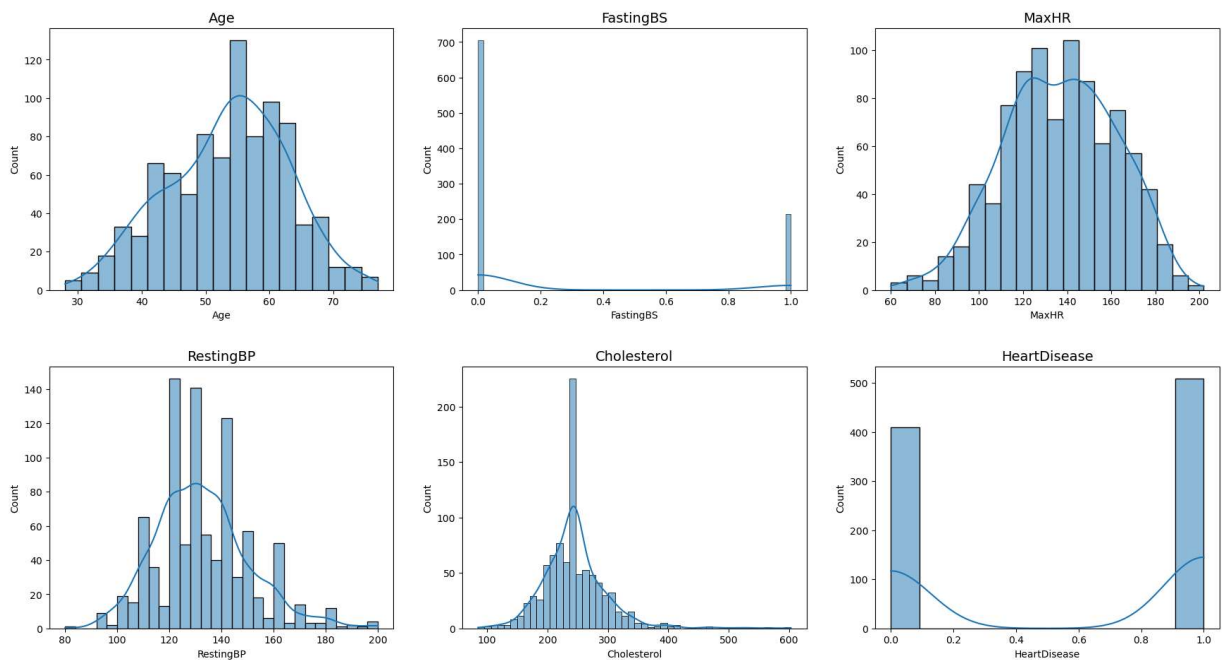
```
In [36]: df['Cholesterol'] = df['Cholesterol'].replace(0 , ch_mean)
df['Cholesterol'] = df['Cholesterol'].round(2)
```

```
In [37]: #checking by smae plotting
plt.figure(figsize=(18, 10))

def plotting(var, num):
    plt.subplot(2, 3, num)
    sns.histplot(df[var], kde=True, bins='auto')
    plt.title(var, fontsize=14)

plotting('Age', 1)
plotting('FastingBS', 2)
plotting('MaxHR', 3)
plotting('RestingBP', 4)
plotting('Cholesterol', 5)
plotting('HeartDisease', 6)

plt.tight_layout(pad=3)
plt.savefig('sub_plot.png' , dpi = 300 , bbox_inches = 'tight')
plt.show()
```



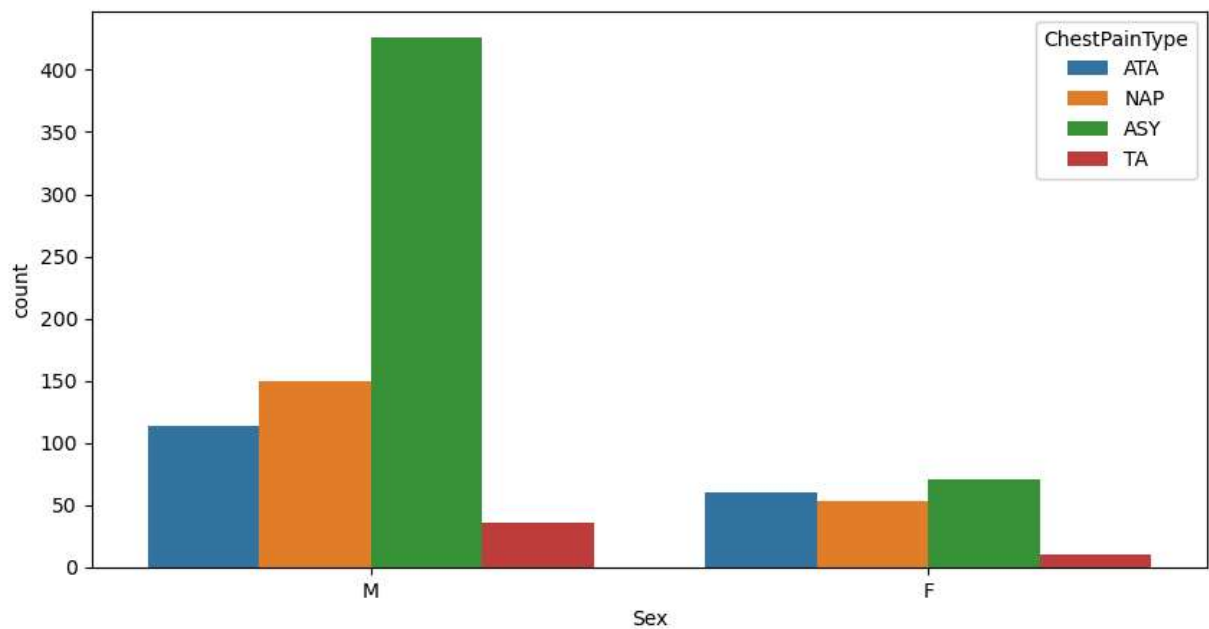
In [38]: `df.head()`

Out[38]:

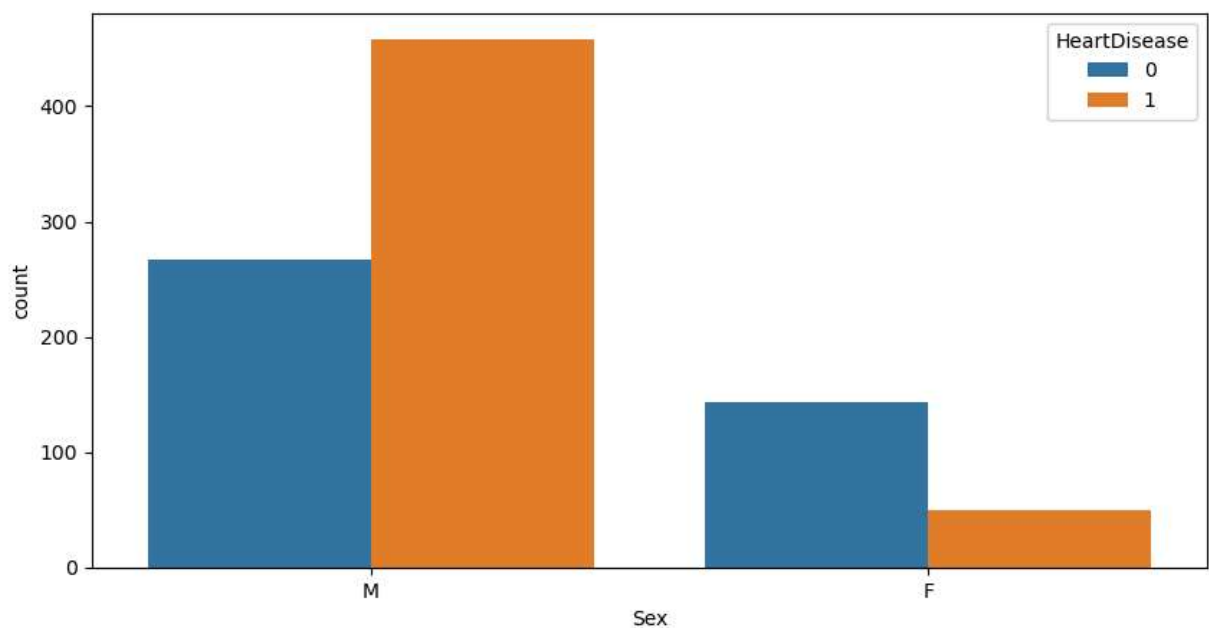
	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina
0	40	M	ATA	140.0	289.0	0	Normal	172	0
1	49	F	NAP	160.0	180.0	0	Normal	156	0
2	37	M	ATA	130.0	283.0	0	ST	98	0
3	48	F	ASY	138.0	214.0	0	Normal	108	0
4	54	M	NAP	150.0	195.0	0	Normal	122	0

In [39]:

```
#graph on based sex
plt.figure(figsize= (10 , 5))
sns.countplot(x = 'Sex' , data = df , hue= 'ChestPainType')
plt.show()
```



```
In [40]: #graph on based sex
plt.figure(figsize= (10 , 5))
sns.countplot(x = 'Sex' , data = df , hue= 'HeartDisease')
plt.show()
```



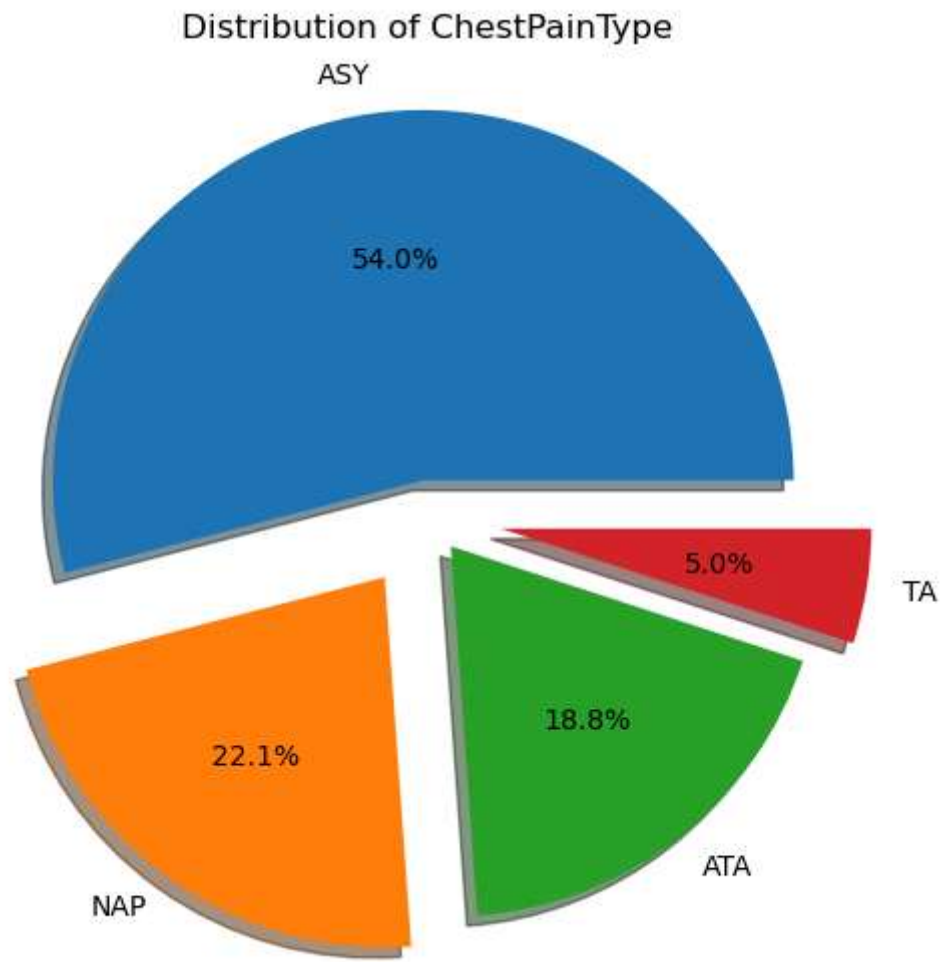
```
In [41]: df['ChestPainType'].value_counts()
```

```
Out[41]: ChestPainType
ASY      496
NAP      203
ATA      173
TA        46
Name: count, dtype: int64
```

```
In [42]: #pic chat of ChestPainType
counts = df['ChestPainType'].value_counts()
```

```
plt.figure(figsize=(10, 6))
plt.pie(counts,
        labels=counts.index,
        autopct='%0.1f%%',
        shadow=True,
        explode=[0.1, 0.2, 0.1, 0.2])

plt.title('Distribution of ChestPainType')
plt.show()
```



```
In [43]: df.head()
```

Out[43]:

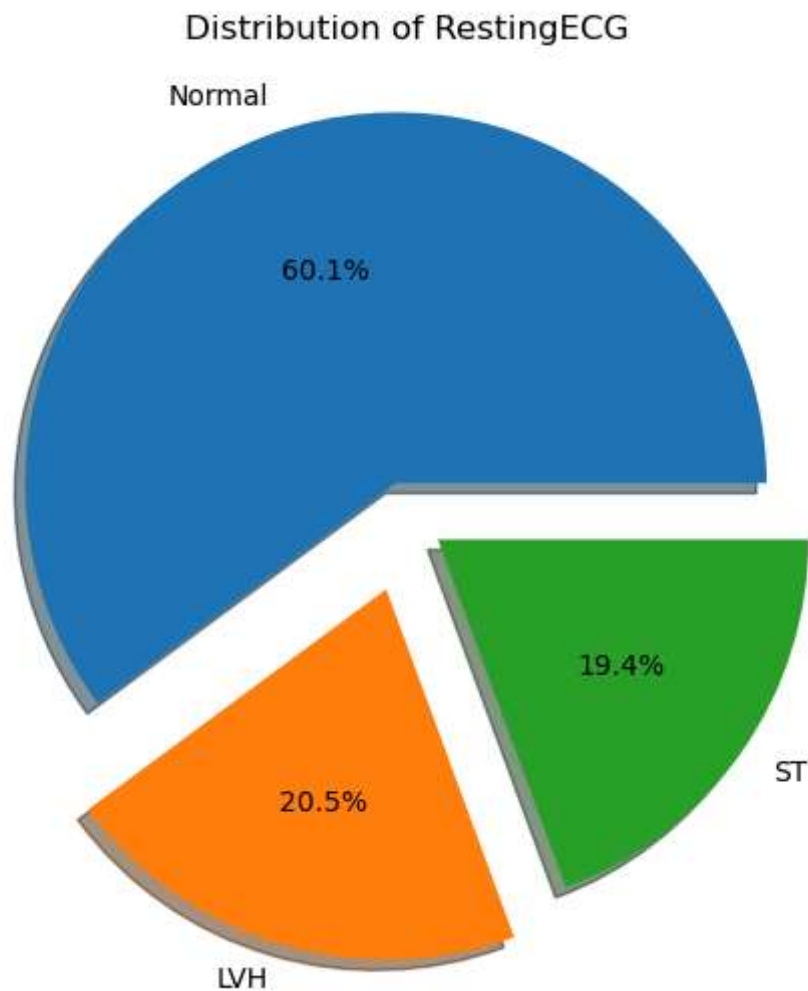
	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAng
0	40	M	ATA	140.0	289.0	0	Normal	172	0
1	49	F	NAP	160.0	180.0	0	Normal	156	0
2	37	M	ATA	130.0	283.0	0	ST	98	0
3	48	F	ASY	138.0	214.0	0	Normal	108	0
4	54	M	NAP	150.0	195.0	0	Normal	122	0



```
In [44]: df['RestingECG'].value_counts()
```

```
Out[44]: RestingECG  
Normal    552  
LVH       188  
ST        178  
Name: count, dtype: int64
```

```
In [45]: ##pic chat of RestingECG  
counts = df['RestingECG'].value_counts()  
  
plt.figure(figsize=(10, 6))  
plt.pie(counts,  
        labels=counts.index,  
        autopct='%0.1f%%',  
        shadow=True,  
        explode=[0.1, 0.2, 0.1])  
  
plt.title('Distribution of RestingECG')  
plt.show()
```



```
In [46]: df.describe()
```

Out[46]:

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	HeartDisease
count	918.000000	918.000000	918.000000	918.000000	918.000000	918.000000
mean	53.510893	132.540893	244.636253	0.233115	136.809368	0.553377
std	9.432617	17.989932	53.318029	0.423046	25.460334	0.497414
min	28.000000	80.000000	85.000000	0.000000	60.000000	0.000000
25%	47.000000	120.000000	214.000000	0.000000	120.000000	0.000000
50%	54.000000	130.000000	244.640000	0.000000	138.000000	1.000000
75%	60.000000	140.000000	267.000000	0.000000	156.000000	1.000000
max	77.000000	200.000000	603.000000	1.000000	202.000000	1.000000

In [47]:

```

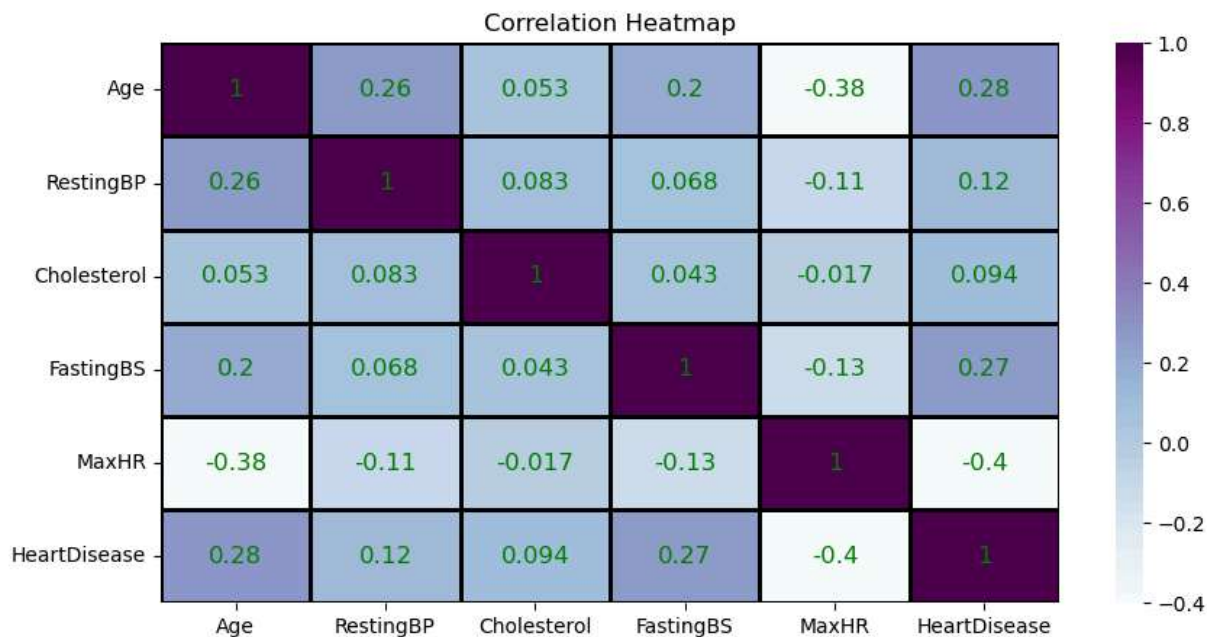
#heat map
plt.figure(figsize=(10 , 5))
col = ['Age', 'RestingBP', 'Cholesterol', 'FastingBS',
       'MaxHR', 'HeartDisease']

v = sns.heatmap(
    df[col].corr(),
    annot=True,
    cmap='BuPu',
    annot_kws={'fontsize': 12, 'color': 'green'},
    linewidths=2,
    linecolor='black',
    xticklabels=True,
    yticklabels=True
)

v.set(title='Correlation Heatmap')

plt.show()

```



In [48]: `df.head()`

Out[48]:

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina
0	40	M	ATA	140.0	289.0	0	Normal	172	0
1	49	F	NAP	160.0	180.0	0	Normal	156	0
2	37	M	ATA	130.0	283.0	0	ST	98	0
3	48	F	ASY	138.0	214.0	0	Normal	108	0
4	54	M	NAP	150.0	195.0	0	Normal	122	0

In [49]:

```
#Label encoding and on-hot encoding
#LB-> sex , FastingBS ,ExerciseAngina
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()

df['Sex'] = le.fit_transform(df['Sex'])
df['RestingECG'] = le.fit_transform(df['RestingECG'])
df['ExerciseAngina'] = le.fit_transform(df['ExerciseAngina'])

#on Hot Encoding 0n -> ChestPainType
df = pd.get_dummies(df, columns=['ChestPainType'], drop_first=True)
```

In [50]: `df.head(10)`

Out[50]:

	Age	Sex	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Hea
0	40	1	140.0	289.0	0	1	172	0	
1	49	0	160.0	180.0	0	1	156	0	
2	37	1	130.0	283.0	0	2	98	0	
3	48	0	138.0	214.0	0	1	108	1	
4	54	1	150.0	195.0	0	1	122	0	
5	39	1	120.0	339.0	0	1	170	0	
6	45	0	130.0	237.0	0	1	170	0	
7	54	1	110.0	208.0	0	1	142	0	
8	37	1	140.0	207.0	0	1	130	1	
9	48	0	120.0	284.0	0	1	120	0	

In [51]: `df.columns`

Out[51]: Index(['Age', 'Sex', 'RestingBP', 'Cholesterol', 'FastingBS', 'RestingECG', 'MaxHR', 'ExerciseAngina', 'HeartDisease', 'ChestPainType_ATA', 'ChestPainType_NAP', 'ChestPainType_TA'], dtype='object')

In [62]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 918 entries, 0 to 917
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Age                    918 non-null   int64
1   Sex                    918 non-null   int64
2   RestingBP              918 non-null   float64
3   Cholesterol             918 non-null   float64
4   FastingBS              918 non-null   int64
5   RestingECG             918 non-null   int64
6   MaxHR                  918 non-null   int64
7   ExerciseAngina         918 non-null   int64
8   HeartDisease           918 non-null   int64
9   ChestPainType_ATA      918 non-null   bool
10  ChestPainType_NAP      918 non-null   bool
11  ChestPainType_TA       918 non-null   bool
dtypes: bool(3), float64(2), int64(7)
memory usage: 67.4 KB
```

In [66]: `#train and test the model`
`from sklearn.model_selection import train_test_split`
`x = df[['Age', 'Sex', 'RestingBP', 'Cholesterol', 'FastingBS', 'RestingECG', 'MaxHR', 'ExerciseAngina', 'ChestPainType_ATA',`

```

        'ChestPainType_NAP', 'ChestPainType_TA']]

y = df['HeartDisease']

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_sta

```

```

In [67]: #Standard Scaling
#SS -> RestingBP , CholesterolMaxHR , RestingECG , MaxHR
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
x_train = scaler.fit_transform(x_train)
x_test = scaler.transform(x_test)

```

```

In [68]: df.head()

```

```

Out[68]:   Age  Sex  RestingBP  Cholesterol  FastingBS  RestingECG  MaxHR  ExerciseAngina  Hea

```

	Age	Sex	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Hea
0	40	1	140.0	289.0	0	1	172	0	
1	49	0	160.0	180.0	0	1	156	0	
2	37	1	130.0	283.0	0	2	98	0	
3	48	0	138.0	214.0	0	1	108	1	
4	54	1	150.0	195.0	0	1	122	0	



```

In [85]: #Logistic Regression
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score , precision_score , recall_score , f1_sc
model = LogisticRegression()
model.fit(x_train , y_train)
L_pred = model.predict(x_test)
print('Logistic Prediction : ' , L_pred[:5])

# for i , val in enumerate(L_pred):
#     if val == 0:
#         print(f"patient{i}: Safe")
#     else:
#         print(f"patient{i}: Not Safe")

for i, (pred, actual) in enumerate(zip(L_pred, y_test), start=1):
    print(f"Patient {i} -> Predicted: {pred}, Actual: {actual}")

```

Logistic Prediction : [0 0 1 1 0]

Patient 1 -> Predicted: 0, Actual: 0
Patient 2 -> Predicted: 0, Actual: 1
Patient 3 -> Predicted: 1, Actual: 1
Patient 4 -> Predicted: 1, Actual: 1
Patient 5 -> Predicted: 0, Actual: 0
Patient 6 -> Predicted: 1, Actual: 1
Patient 7 -> Predicted: 0, Actual: 1
Patient 8 -> Predicted: 0, Actual: 0
Patient 9 -> Predicted: 0, Actual: 1
Patient 10 -> Predicted: 1, Actual: 1
Patient 11 -> Predicted: 0, Actual: 0
Patient 12 -> Predicted: 1, Actual: 0
Patient 13 -> Predicted: 0, Actual: 0
Patient 14 -> Predicted: 0, Actual: 0
Patient 15 -> Predicted: 1, Actual: 1
Patient 16 -> Predicted: 1, Actual: 1
Patient 17 -> Predicted: 0, Actual: 0
Patient 18 -> Predicted: 1, Actual: 1
Patient 19 -> Predicted: 0, Actual: 1
Patient 20 -> Predicted: 0, Actual: 0
Patient 21 -> Predicted: 1, Actual: 1
Patient 22 -> Predicted: 1, Actual: 0
Patient 23 -> Predicted: 0, Actual: 0
Patient 24 -> Predicted: 1, Actual: 0
Patient 25 -> Predicted: 1, Actual: 1
Patient 26 -> Predicted: 1, Actual: 1
Patient 27 -> Predicted: 0, Actual: 0
Patient 28 -> Predicted: 1, Actual: 1
Patient 29 -> Predicted: 0, Actual: 0
Patient 30 -> Predicted: 0, Actual: 0
Patient 31 -> Predicted: 0, Actual: 1
Patient 32 -> Predicted: 0, Actual: 1
Patient 33 -> Predicted: 1, Actual: 0
Patient 34 -> Predicted: 1, Actual: 1
Patient 35 -> Predicted: 1, Actual: 1
Patient 36 -> Predicted: 0, Actual: 0
Patient 37 -> Predicted: 1, Actual: 1
Patient 38 -> Predicted: 0, Actual: 0
Patient 39 -> Predicted: 1, Actual: 1
Patient 40 -> Predicted: 0, Actual: 1
Patient 41 -> Predicted: 1, Actual: 1
Patient 42 -> Predicted: 0, Actual: 1
Patient 43 -> Predicted: 0, Actual: 1
Patient 44 -> Predicted: 0, Actual: 0
Patient 45 -> Predicted: 0, Actual: 0
Patient 46 -> Predicted: 0, Actual: 0
Patient 47 -> Predicted: 1, Actual: 1
Patient 48 -> Predicted: 1, Actual: 1
Patient 49 -> Predicted: 1, Actual: 1
Patient 50 -> Predicted: 1, Actual: 0
Patient 51 -> Predicted: 0, Actual: 1
Patient 52 -> Predicted: 0, Actual: 0
Patient 53 -> Predicted: 1, Actual: 1
Patient 54 -> Predicted: 1, Actual: 1
Patient 55 -> Predicted: 1, Actual: 1

Patient 56 -> Predicted: 1, Actual: 0
Patient 57 -> Predicted: 0, Actual: 0
Patient 58 -> Predicted: 1, Actual: 1
Patient 59 -> Predicted: 1, Actual: 1
Patient 60 -> Predicted: 1, Actual: 1
Patient 61 -> Predicted: 0, Actual: 0
Patient 62 -> Predicted: 0, Actual: 0
Patient 63 -> Predicted: 0, Actual: 0
Patient 64 -> Predicted: 1, Actual: 1
Patient 65 -> Predicted: 0, Actual: 0
Patient 66 -> Predicted: 1, Actual: 1
Patient 67 -> Predicted: 1, Actual: 1
Patient 68 -> Predicted: 1, Actual: 1
Patient 69 -> Predicted: 1, Actual: 1
Patient 70 -> Predicted: 0, Actual: 1
Patient 71 -> Predicted: 1, Actual: 0
Patient 72 -> Predicted: 0, Actual: 0
Patient 73 -> Predicted: 1, Actual: 1
Patient 74 -> Predicted: 0, Actual: 1
Patient 75 -> Predicted: 0, Actual: 0
Patient 76 -> Predicted: 1, Actual: 1
Patient 77 -> Predicted: 1, Actual: 0
Patient 78 -> Predicted: 0, Actual: 1
Patient 79 -> Predicted: 0, Actual: 0
Patient 80 -> Predicted: 1, Actual: 1
Patient 81 -> Predicted: 1, Actual: 1
Patient 82 -> Predicted: 1, Actual: 1
Patient 83 -> Predicted: 1, Actual: 1
Patient 84 -> Predicted: 0, Actual: 0
Patient 85 -> Predicted: 1, Actual: 1
Patient 86 -> Predicted: 0, Actual: 1
Patient 87 -> Predicted: 0, Actual: 0
Patient 88 -> Predicted: 1, Actual: 1
Patient 89 -> Predicted: 0, Actual: 0
Patient 90 -> Predicted: 0, Actual: 0
Patient 91 -> Predicted: 0, Actual: 1
Patient 92 -> Predicted: 1, Actual: 1
Patient 93 -> Predicted: 1, Actual: 1
Patient 94 -> Predicted: 0, Actual: 0
Patient 95 -> Predicted: 0, Actual: 0
Patient 96 -> Predicted: 0, Actual: 0
Patient 97 -> Predicted: 1, Actual: 1
Patient 98 -> Predicted: 0, Actual: 0
Patient 99 -> Predicted: 1, Actual: 1
Patient 100 -> Predicted: 1, Actual: 1
Patient 101 -> Predicted: 0, Actual: 0
Patient 102 -> Predicted: 1, Actual: 1
Patient 103 -> Predicted: 1, Actual: 1
Patient 104 -> Predicted: 0, Actual: 0
Patient 105 -> Predicted: 1, Actual: 1
Patient 106 -> Predicted: 1, Actual: 1
Patient 107 -> Predicted: 0, Actual: 0
Patient 108 -> Predicted: 1, Actual: 1
Patient 109 -> Predicted: 1, Actual: 1
Patient 110 -> Predicted: 1, Actual: 0
Patient 111 -> Predicted: 1, Actual: 1

Patient 112 -> Predicted: 1, Actual: 1
Patient 113 -> Predicted: 0, Actual: 0
Patient 114 -> Predicted: 0, Actual: 1
Patient 115 -> Predicted: 1, Actual: 1
Patient 116 -> Predicted: 0, Actual: 0
Patient 117 -> Predicted: 1, Actual: 1
Patient 118 -> Predicted: 0, Actual: 0
Patient 119 -> Predicted: 1, Actual: 1
Patient 120 -> Predicted: 0, Actual: 0
Patient 121 -> Predicted: 1, Actual: 1
Patient 122 -> Predicted: 1, Actual: 1
Patient 123 -> Predicted: 1, Actual: 1
Patient 124 -> Predicted: 1, Actual: 1
Patient 125 -> Predicted: 0, Actual: 0
Patient 126 -> Predicted: 0, Actual: 0
Patient 127 -> Predicted: 0, Actual: 0
Patient 128 -> Predicted: 0, Actual: 1
Patient 129 -> Predicted: 0, Actual: 0
Patient 130 -> Predicted: 1, Actual: 1
Patient 131 -> Predicted: 0, Actual: 0
Patient 132 -> Predicted: 1, Actual: 1
Patient 133 -> Predicted: 1, Actual: 1
Patient 134 -> Predicted: 1, Actual: 1
Patient 135 -> Predicted: 1, Actual: 1
Patient 136 -> Predicted: 0, Actual: 0
Patient 137 -> Predicted: 1, Actual: 1
Patient 138 -> Predicted: 0, Actual: 1
Patient 139 -> Predicted: 0, Actual: 0
Patient 140 -> Predicted: 1, Actual: 1
Patient 141 -> Predicted: 1, Actual: 0
Patient 142 -> Predicted: 0, Actual: 0
Patient 143 -> Predicted: 1, Actual: 1
Patient 144 -> Predicted: 1, Actual: 1
Patient 145 -> Predicted: 1, Actual: 1
Patient 146 -> Predicted: 1, Actual: 0
Patient 147 -> Predicted: 0, Actual: 1
Patient 148 -> Predicted: 0, Actual: 0
Patient 149 -> Predicted: 1, Actual: 0
Patient 150 -> Predicted: 1, Actual: 1
Patient 151 -> Predicted: 0, Actual: 0
Patient 152 -> Predicted: 1, Actual: 1
Patient 153 -> Predicted: 1, Actual: 1
Patient 154 -> Predicted: 0, Actual: 0
Patient 155 -> Predicted: 0, Actual: 1
Patient 156 -> Predicted: 0, Actual: 0
Patient 157 -> Predicted: 0, Actual: 1
Patient 158 -> Predicted: 0, Actual: 0
Patient 159 -> Predicted: 0, Actual: 0
Patient 160 -> Predicted: 1, Actual: 1
Patient 161 -> Predicted: 0, Actual: 0
Patient 162 -> Predicted: 0, Actual: 0
Patient 163 -> Predicted: 1, Actual: 1
Patient 164 -> Predicted: 0, Actual: 0
Patient 165 -> Predicted: 1, Actual: 1
Patient 166 -> Predicted: 1, Actual: 1
Patient 167 -> Predicted: 1, Actual: 1

Patient 168 -> Predicted: 1, Actual: 1
 Patient 169 -> Predicted: 0, Actual: 0
 Patient 170 -> Predicted: 0, Actual: 0
 Patient 171 -> Predicted: 1, Actual: 1
 Patient 172 -> Predicted: 1, Actual: 1
 Patient 173 -> Predicted: 0, Actual: 0
 Patient 174 -> Predicted: 1, Actual: 0
 Patient 175 -> Predicted: 0, Actual: 0
 Patient 176 -> Predicted: 1, Actual: 1
 Patient 177 -> Predicted: 1, Actual: 1
 Patient 178 -> Predicted: 1, Actual: 1
 Patient 179 -> Predicted: 0, Actual: 0
 Patient 180 -> Predicted: 0, Actual: 1
 Patient 181 -> Predicted: 1, Actual: 1
 Patient 182 -> Predicted: 1, Actual: 1
 Patient 183 -> Predicted: 0, Actual: 0
 Patient 184 -> Predicted: 1, Actual: 1

```

In [84]: #find matrices
from sklearn.metrics import accuracy_score , precision_score , recall_score , f1_score
acc = accuracy_score(y_test , L_pred)
print('Accuracy Score : ' , round(acc , 2))

pre = precision_score(y_test , L_pred)
print('Precision Score : ' , round(pre , 2))

re = recall_score(y_test , L_pred)
print('Recall Score : ' , round(re , 2))

f1 = f1_score(y_test , L_pred)
print('F1 Score : ' , round(f1 , 2) )
  
```

Accuracy Score : 0.81
 Precision Score : 0.87
 Recall Score : 0.79
 F1 Score : 0.83

```

In [127... #Another Model test -> RandomForestClassifier
from sklearn.ensemble import RandomForestClassifier
model = RandomForestClassifier()
model.fit(x_train , y_train)
R_pred = model.predict(x_test)

print('Forest Predicated : ' , R_pred[:5])

for i ,(pred , actual) in enumerate(zip(R_pred , y_test) , start = 1):
    print(f"Patient {i} -> Predict: {pred} , Actual: {actual}")
  
```

Forest Predicated : [0 0 1 1 0]

Patient 1	-> Predict: 0	, Actual: 0
Patient 2	-> Predict: 0	, Actual: 1
Patient 3	-> Predict: 1	, Actual: 1
Patient 4	-> Predict: 1	, Actual: 1
Patient 5	-> Predict: 0	, Actual: 0
Patient 6	-> Predict: 1	, Actual: 1
Patient 7	-> Predict: 1	, Actual: 1
Patient 8	-> Predict: 1	, Actual: 0
Patient 9	-> Predict: 0	, Actual: 1
Patient 10	-> Predict: 1	, Actual: 1
Patient 11	-> Predict: 0	, Actual: 0
Patient 12	-> Predict: 0	, Actual: 0
Patient 13	-> Predict: 0	, Actual: 0
Patient 14	-> Predict: 0	, Actual: 0
Patient 15	-> Predict: 1	, Actual: 1
Patient 16	-> Predict: 1	, Actual: 1
Patient 17	-> Predict: 0	, Actual: 0
Patient 18	-> Predict: 1	, Actual: 1
Patient 19	-> Predict: 0	, Actual: 1
Patient 20	-> Predict: 0	, Actual: 0
Patient 21	-> Predict: 1	, Actual: 1
Patient 22	-> Predict: 1	, Actual: 0
Patient 23	-> Predict: 0	, Actual: 0
Patient 24	-> Predict: 0	, Actual: 0
Patient 25	-> Predict: 1	, Actual: 1
Patient 26	-> Predict: 1	, Actual: 1
Patient 27	-> Predict: 0	, Actual: 0
Patient 28	-> Predict: 1	, Actual: 1
Patient 29	-> Predict: 0	, Actual: 0
Patient 30	-> Predict: 0	, Actual: 0
Patient 31	-> Predict: 0	, Actual: 1
Patient 32	-> Predict: 0	, Actual: 1
Patient 33	-> Predict: 0	, Actual: 0
Patient 34	-> Predict: 1	, Actual: 1
Patient 35	-> Predict: 1	, Actual: 1
Patient 36	-> Predict: 0	, Actual: 0
Patient 37	-> Predict: 1	, Actual: 1
Patient 38	-> Predict: 0	, Actual: 0
Patient 39	-> Predict: 1	, Actual: 1
Patient 40	-> Predict: 1	, Actual: 1
Patient 41	-> Predict: 1	, Actual: 1
Patient 42	-> Predict: 1	, Actual: 1
Patient 43	-> Predict: 1	, Actual: 1
Patient 44	-> Predict: 0	, Actual: 0
Patient 45	-> Predict: 1	, Actual: 0
Patient 46	-> Predict: 1	, Actual: 0
Patient 47	-> Predict: 0	, Actual: 1
Patient 48	-> Predict: 1	, Actual: 1
Patient 49	-> Predict: 1	, Actual: 1
Patient 50	-> Predict: 1	, Actual: 0
Patient 51	-> Predict: 0	, Actual: 1
Patient 52	-> Predict: 0	, Actual: 0
Patient 53	-> Predict: 1	, Actual: 1
Patient 54	-> Predict: 1	, Actual: 1
Patient 55	-> Predict: 1	, Actual: 1

Patient 56 -> Predict: 1 , Actual: 0
Patient 57 -> Predict: 0 , Actual: 0
Patient 58 -> Predict: 1 , Actual: 1
Patient 59 -> Predict: 1 , Actual: 1
Patient 60 -> Predict: 1 , Actual: 1
Patient 61 -> Predict: 0 , Actual: 0
Patient 62 -> Predict: 1 , Actual: 0
Patient 63 -> Predict: 0 , Actual: 0
Patient 64 -> Predict: 1 , Actual: 1
Patient 65 -> Predict: 0 , Actual: 0
Patient 66 -> Predict: 1 , Actual: 1
Patient 67 -> Predict: 1 , Actual: 1
Patient 68 -> Predict: 1 , Actual: 1
Patient 69 -> Predict: 1 , Actual: 1
Patient 70 -> Predict: 0 , Actual: 1
Patient 71 -> Predict: 1 , Actual: 0
Patient 72 -> Predict: 0 , Actual: 0
Patient 73 -> Predict: 1 , Actual: 1
Patient 74 -> Predict: 0 , Actual: 1
Patient 75 -> Predict: 0 , Actual: 0
Patient 76 -> Predict: 0 , Actual: 1
Patient 77 -> Predict: 1 , Actual: 0
Patient 78 -> Predict: 1 , Actual: 1
Patient 79 -> Predict: 0 , Actual: 0
Patient 80 -> Predict: 1 , Actual: 1
Patient 81 -> Predict: 1 , Actual: 1
Patient 82 -> Predict: 1 , Actual: 1
Patient 83 -> Predict: 1 , Actual: 1
Patient 84 -> Predict: 0 , Actual: 0
Patient 85 -> Predict: 1 , Actual: 1
Patient 86 -> Predict: 1 , Actual: 1
Patient 87 -> Predict: 0 , Actual: 0
Patient 88 -> Predict: 1 , Actual: 1
Patient 89 -> Predict: 0 , Actual: 0
Patient 90 -> Predict: 0 , Actual: 0
Patient 91 -> Predict: 0 , Actual: 1
Patient 92 -> Predict: 1 , Actual: 1
Patient 93 -> Predict: 1 , Actual: 1
Patient 94 -> Predict: 0 , Actual: 0
Patient 95 -> Predict: 0 , Actual: 0
Patient 96 -> Predict: 0 , Actual: 0
Patient 97 -> Predict: 1 , Actual: 1
Patient 98 -> Predict: 1 , Actual: 0
Patient 99 -> Predict: 1 , Actual: 1
Patient 100 -> Predict: 1 , Actual: 1
Patient 101 -> Predict: 0 , Actual: 0
Patient 102 -> Predict: 1 , Actual: 1
Patient 103 -> Predict: 1 , Actual: 1
Patient 104 -> Predict: 0 , Actual: 0
Patient 105 -> Predict: 1 , Actual: 1
Patient 106 -> Predict: 1 , Actual: 1
Patient 107 -> Predict: 0 , Actual: 0
Patient 108 -> Predict: 1 , Actual: 1
Patient 109 -> Predict: 1 , Actual: 1
Patient 110 -> Predict: 1 , Actual: 0
Patient 111 -> Predict: 0 , Actual: 1

Patient 112 -> Predict: 1 , Actual: 1
Patient 113 -> Predict: 0 , Actual: 0
Patient 114 -> Predict: 0 , Actual: 1
Patient 115 -> Predict: 1 , Actual: 1
Patient 116 -> Predict: 0 , Actual: 0
Patient 117 -> Predict: 1 , Actual: 1
Patient 118 -> Predict: 0 , Actual: 0
Patient 119 -> Predict: 1 , Actual: 1
Patient 120 -> Predict: 0 , Actual: 0
Patient 121 -> Predict: 1 , Actual: 1
Patient 122 -> Predict: 1 , Actual: 1
Patient 123 -> Predict: 1 , Actual: 1
Patient 124 -> Predict: 1 , Actual: 1
Patient 125 -> Predict: 0 , Actual: 0
Patient 126 -> Predict: 0 , Actual: 0
Patient 127 -> Predict: 0 , Actual: 0
Patient 128 -> Predict: 0 , Actual: 1
Patient 129 -> Predict: 0 , Actual: 0
Patient 130 -> Predict: 0 , Actual: 1
Patient 131 -> Predict: 1 , Actual: 0
Patient 132 -> Predict: 1 , Actual: 1
Patient 133 -> Predict: 1 , Actual: 1
Patient 134 -> Predict: 1 , Actual: 1
Patient 135 -> Predict: 1 , Actual: 1
Patient 136 -> Predict: 0 , Actual: 0
Patient 137 -> Predict: 1 , Actual: 1
Patient 138 -> Predict: 0 , Actual: 1
Patient 139 -> Predict: 0 , Actual: 0
Patient 140 -> Predict: 1 , Actual: 1
Patient 141 -> Predict: 0 , Actual: 0
Patient 142 -> Predict: 0 , Actual: 0
Patient 143 -> Predict: 1 , Actual: 1
Patient 144 -> Predict: 1 , Actual: 1
Patient 145 -> Predict: 1 , Actual: 1
Patient 146 -> Predict: 1 , Actual: 0
Patient 147 -> Predict: 0 , Actual: 1
Patient 148 -> Predict: 0 , Actual: 0
Patient 149 -> Predict: 1 , Actual: 0
Patient 150 -> Predict: 1 , Actual: 1
Patient 151 -> Predict: 0 , Actual: 0
Patient 152 -> Predict: 1 , Actual: 1
Patient 153 -> Predict: 1 , Actual: 1
Patient 154 -> Predict: 0 , Actual: 0
Patient 155 -> Predict: 0 , Actual: 1
Patient 156 -> Predict: 0 , Actual: 0
Patient 157 -> Predict: 0 , Actual: 1
Patient 158 -> Predict: 0 , Actual: 0
Patient 159 -> Predict: 0 , Actual: 0
Patient 160 -> Predict: 1 , Actual: 1
Patient 161 -> Predict: 0 , Actual: 0
Patient 162 -> Predict: 0 , Actual: 0
Patient 163 -> Predict: 1 , Actual: 1
Patient 164 -> Predict: 0 , Actual: 0
Patient 165 -> Predict: 1 , Actual: 1
Patient 166 -> Predict: 1 , Actual: 1
Patient 167 -> Predict: 1 , Actual: 1

```

Patient 168 -> Predict: 1 , Actual: 1
Patient 169 -> Predict: 1 , Actual: 0
Patient 170 -> Predict: 1 , Actual: 0
Patient 171 -> Predict: 1 , Actual: 1
Patient 172 -> Predict: 1 , Actual: 1
Patient 173 -> Predict: 0 , Actual: 0
Patient 174 -> Predict: 1 , Actual: 0
Patient 175 -> Predict: 0 , Actual: 0
Patient 176 -> Predict: 1 , Actual: 1
Patient 177 -> Predict: 1 , Actual: 1
Patient 178 -> Predict: 1 , Actual: 1
Patient 179 -> Predict: 0 , Actual: 0
Patient 180 -> Predict: 1 , Actual: 1
Patient 181 -> Predict: 0 , Actual: 1
Patient 182 -> Predict: 1 , Actual: 1
Patient 183 -> Predict: 0 , Actual: 0
Patient 184 -> Predict: 1 , Actual: 1

```

```

In [89]: #find matrices of RandomForestClassifier
from sklearn.metrics import accuracy_score , precision_score , recall_score , f1_score
acc = accuracy_score(y_test , R_pred)
print('Accuracy Score : ' , round(acc , 2))

pre = precision_score(y_test , R_pred)
print('Precision Score : ' , round(pre , 2))

re = recall_score(y_test , R_pred)
print('Recall Score : ' , round(re , 2))

f1 = f1_score(y_test , R_pred)
print('F1 Score : ' , round(f1 , 2) )

```

```

Accuracy Score : 0.78
Precision Score : 0.83
Recall Score : 0.79
F1 Score : 0.81

```

```

In [95]: # Another Model -> XGBClassifier
from xgboost import XGBClassifier
model = XGBClassifier()
model.fit(x_train , y_train)
X_pred = model.predict(x_test)

print('Forest Predicated : ' , X_pred[:5])

for i ,(pred , actual) in enumerate(zip(X_pred , y_test) , start = 1):
    print(f"Patient: {i} -> Predict: {pred} , Actual: {actual}")

```

Forest Predicated : [0 1 1 1 0]

Patient: 1	-> Predict: 0	, Actual: 0
Patient: 2	-> Predict: 1	, Actual: 1
Patient: 3	-> Predict: 1	, Actual: 1
Patient: 4	-> Predict: 1	, Actual: 1
Patient: 5	-> Predict: 0	, Actual: 0
Patient: 6	-> Predict: 0	, Actual: 1
Patient: 7	-> Predict: 1	, Actual: 1
Patient: 8	-> Predict: 1	, Actual: 0
Patient: 9	-> Predict: 0	, Actual: 1
Patient: 10	-> Predict: 1	, Actual: 1
Patient: 11	-> Predict: 0	, Actual: 0
Patient: 12	-> Predict: 0	, Actual: 0
Patient: 13	-> Predict: 0	, Actual: 0
Patient: 14	-> Predict: 0	, Actual: 0
Patient: 15	-> Predict: 1	, Actual: 1
Patient: 16	-> Predict: 1	, Actual: 1
Patient: 17	-> Predict: 0	, Actual: 0
Patient: 18	-> Predict: 1	, Actual: 1
Patient: 19	-> Predict: 0	, Actual: 1
Patient: 20	-> Predict: 0	, Actual: 0
Patient: 21	-> Predict: 1	, Actual: 1
Patient: 22	-> Predict: 1	, Actual: 0
Patient: 23	-> Predict: 0	, Actual: 0
Patient: 24	-> Predict: 0	, Actual: 0
Patient: 25	-> Predict: 1	, Actual: 1
Patient: 26	-> Predict: 1	, Actual: 1
Patient: 27	-> Predict: 0	, Actual: 0
Patient: 28	-> Predict: 1	, Actual: 1
Patient: 29	-> Predict: 0	, Actual: 0
Patient: 30	-> Predict: 0	, Actual: 0
Patient: 31	-> Predict: 0	, Actual: 1
Patient: 32	-> Predict: 0	, Actual: 1
Patient: 33	-> Predict: 1	, Actual: 0
Patient: 34	-> Predict: 1	, Actual: 1
Patient: 35	-> Predict: 1	, Actual: 1
Patient: 36	-> Predict: 0	, Actual: 0
Patient: 37	-> Predict: 1	, Actual: 1
Patient: 38	-> Predict: 0	, Actual: 0
Patient: 39	-> Predict: 1	, Actual: 1
Patient: 40	-> Predict: 1	, Actual: 1
Patient: 41	-> Predict: 0	, Actual: 1
Patient: 42	-> Predict: 1	, Actual: 1
Patient: 43	-> Predict: 0	, Actual: 1
Patient: 44	-> Predict: 0	, Actual: 0
Patient: 45	-> Predict: 1	, Actual: 0
Patient: 46	-> Predict: 1	, Actual: 0
Patient: 47	-> Predict: 0	, Actual: 1
Patient: 48	-> Predict: 1	, Actual: 1
Patient: 49	-> Predict: 1	, Actual: 1
Patient: 50	-> Predict: 1	, Actual: 0
Patient: 51	-> Predict: 0	, Actual: 1
Patient: 52	-> Predict: 0	, Actual: 0
Patient: 53	-> Predict: 1	, Actual: 1
Patient: 54	-> Predict: 1	, Actual: 1
Patient: 55	-> Predict: 1	, Actual: 1

Patient: 56 -> Predict: 0 , Actual: 0
Patient: 57 -> Predict: 0 , Actual: 0
Patient: 58 -> Predict: 1 , Actual: 1
Patient: 59 -> Predict: 1 , Actual: 1
Patient: 60 -> Predict: 1 , Actual: 1
Patient: 61 -> Predict: 0 , Actual: 0
Patient: 62 -> Predict: 1 , Actual: 0
Patient: 63 -> Predict: 0 , Actual: 0
Patient: 64 -> Predict: 1 , Actual: 1
Patient: 65 -> Predict: 0 , Actual: 0
Patient: 66 -> Predict: 1 , Actual: 1
Patient: 67 -> Predict: 1 , Actual: 1
Patient: 68 -> Predict: 1 , Actual: 1
Patient: 69 -> Predict: 1 , Actual: 1
Patient: 70 -> Predict: 0 , Actual: 1
Patient: 71 -> Predict: 1 , Actual: 0
Patient: 72 -> Predict: 0 , Actual: 0
Patient: 73 -> Predict: 1 , Actual: 1
Patient: 74 -> Predict: 0 , Actual: 1
Patient: 75 -> Predict: 0 , Actual: 0
Patient: 76 -> Predict: 1 , Actual: 1
Patient: 77 -> Predict: 1 , Actual: 0
Patient: 78 -> Predict: 0 , Actual: 1
Patient: 79 -> Predict: 1 , Actual: 0
Patient: 80 -> Predict: 1 , Actual: 1
Patient: 81 -> Predict: 1 , Actual: 1
Patient: 82 -> Predict: 1 , Actual: 1
Patient: 83 -> Predict: 1 , Actual: 1
Patient: 84 -> Predict: 0 , Actual: 0
Patient: 85 -> Predict: 1 , Actual: 1
Patient: 86 -> Predict: 1 , Actual: 1
Patient: 87 -> Predict: 0 , Actual: 0
Patient: 88 -> Predict: 1 , Actual: 1
Patient: 89 -> Predict: 0 , Actual: 0
Patient: 90 -> Predict: 0 , Actual: 0
Patient: 91 -> Predict: 0 , Actual: 1
Patient: 92 -> Predict: 1 , Actual: 1
Patient: 93 -> Predict: 1 , Actual: 1
Patient: 94 -> Predict: 0 , Actual: 0
Patient: 95 -> Predict: 1 , Actual: 0
Patient: 96 -> Predict: 0 , Actual: 0
Patient: 97 -> Predict: 1 , Actual: 1
Patient: 98 -> Predict: 1 , Actual: 0
Patient: 99 -> Predict: 1 , Actual: 1
Patient: 100 -> Predict: 1 , Actual: 1
Patient: 101 -> Predict: 0 , Actual: 0
Patient: 102 -> Predict: 1 , Actual: 1
Patient: 103 -> Predict: 1 , Actual: 1
Patient: 104 -> Predict: 1 , Actual: 0
Patient: 105 -> Predict: 1 , Actual: 1
Patient: 106 -> Predict: 1 , Actual: 1
Patient: 107 -> Predict: 0 , Actual: 0
Patient: 108 -> Predict: 1 , Actual: 1
Patient: 109 -> Predict: 1 , Actual: 1
Patient: 110 -> Predict: 1 , Actual: 0
Patient: 111 -> Predict: 0 , Actual: 1

Patient: 112 -> Predict: 1 , Actual: 1
Patient: 113 -> Predict: 0 , Actual: 0
Patient: 114 -> Predict: 0 , Actual: 1
Patient: 115 -> Predict: 1 , Actual: 1
Patient: 116 -> Predict: 0 , Actual: 0
Patient: 117 -> Predict: 1 , Actual: 1
Patient: 118 -> Predict: 0 , Actual: 0
Patient: 119 -> Predict: 1 , Actual: 1
Patient: 120 -> Predict: 0 , Actual: 0
Patient: 121 -> Predict: 1 , Actual: 1
Patient: 122 -> Predict: 1 , Actual: 1
Patient: 123 -> Predict: 1 , Actual: 1
Patient: 124 -> Predict: 1 , Actual: 1
Patient: 125 -> Predict: 0 , Actual: 0
Patient: 126 -> Predict: 0 , Actual: 0
Patient: 127 -> Predict: 0 , Actual: 0
Patient: 128 -> Predict: 0 , Actual: 1
Patient: 129 -> Predict: 0 , Actual: 0
Patient: 130 -> Predict: 1 , Actual: 1
Patient: 131 -> Predict: 1 , Actual: 0
Patient: 132 -> Predict: 1 , Actual: 1
Patient: 133 -> Predict: 1 , Actual: 1
Patient: 134 -> Predict: 0 , Actual: 1
Patient: 135 -> Predict: 1 , Actual: 1
Patient: 136 -> Predict: 0 , Actual: 0
Patient: 137 -> Predict: 1 , Actual: 1
Patient: 138 -> Predict: 0 , Actual: 1
Patient: 139 -> Predict: 0 , Actual: 0
Patient: 140 -> Predict: 1 , Actual: 1
Patient: 141 -> Predict: 0 , Actual: 0
Patient: 142 -> Predict: 0 , Actual: 0
Patient: 143 -> Predict: 1 , Actual: 1
Patient: 144 -> Predict: 1 , Actual: 1
Patient: 145 -> Predict: 1 , Actual: 1
Patient: 146 -> Predict: 1 , Actual: 0
Patient: 147 -> Predict: 1 , Actual: 1
Patient: 148 -> Predict: 0 , Actual: 0
Patient: 149 -> Predict: 1 , Actual: 0
Patient: 150 -> Predict: 1 , Actual: 1
Patient: 151 -> Predict: 0 , Actual: 0
Patient: 152 -> Predict: 1 , Actual: 1
Patient: 153 -> Predict: 1 , Actual: 1
Patient: 154 -> Predict: 0 , Actual: 0
Patient: 155 -> Predict: 0 , Actual: 1
Patient: 156 -> Predict: 1 , Actual: 0
Patient: 157 -> Predict: 0 , Actual: 1
Patient: 158 -> Predict: 0 , Actual: 0
Patient: 159 -> Predict: 0 , Actual: 0
Patient: 160 -> Predict: 1 , Actual: 1
Patient: 161 -> Predict: 0 , Actual: 0
Patient: 162 -> Predict: 0 , Actual: 0
Patient: 163 -> Predict: 1 , Actual: 1
Patient: 164 -> Predict: 0 , Actual: 0
Patient: 165 -> Predict: 1 , Actual: 1
Patient: 166 -> Predict: 1 , Actual: 1
Patient: 167 -> Predict: 1 , Actual: 1

Patient: 168 -> Predict: 1 , Actual: 1
 Patient: 169 -> Predict: 0 , Actual: 0
 Patient: 170 -> Predict: 1 , Actual: 0
 Patient: 171 -> Predict: 1 , Actual: 1
 Patient: 172 -> Predict: 0 , Actual: 1
 Patient: 173 -> Predict: 0 , Actual: 0
 Patient: 174 -> Predict: 1 , Actual: 0
 Patient: 175 -> Predict: 0 , Actual: 0
 Patient: 176 -> Predict: 1 , Actual: 1
 Patient: 177 -> Predict: 1 , Actual: 1
 Patient: 178 -> Predict: 1 , Actual: 1
 Patient: 179 -> Predict: 0 , Actual: 0
 Patient: 180 -> Predict: 1 , Actual: 1
 Patient: 181 -> Predict: 0 , Actual: 1
 Patient: 182 -> Predict: 1 , Actual: 1
 Patient: 183 -> Predict: 0 , Actual: 0
 Patient: 184 -> Predict: 1 , Actual: 1

```

In [96]: #find matrices of XGBclassifier
from sklearn.metrics import accuracy_score , precision_score , recall_score , f1_score
acc = accuracy_score(y_test , X_pred)
print('Accuracy Score : ' , round(acc , 2))

pre = precision_score(y_test , X_pred)
print('Precision Score : ' , round(pre , 2))

re = recall_score(y_test , X_pred)
print('Recall Score : ' , round(re , 2))

f1 = f1_score(y_test , X_pred)
print('F1 Score : ' , round(f1 , 2) )
  
```

Accuracy Score : 0.77
 Precision Score : 0.81
 Recall Score : 0.79
 F1 Score : 0.8

```

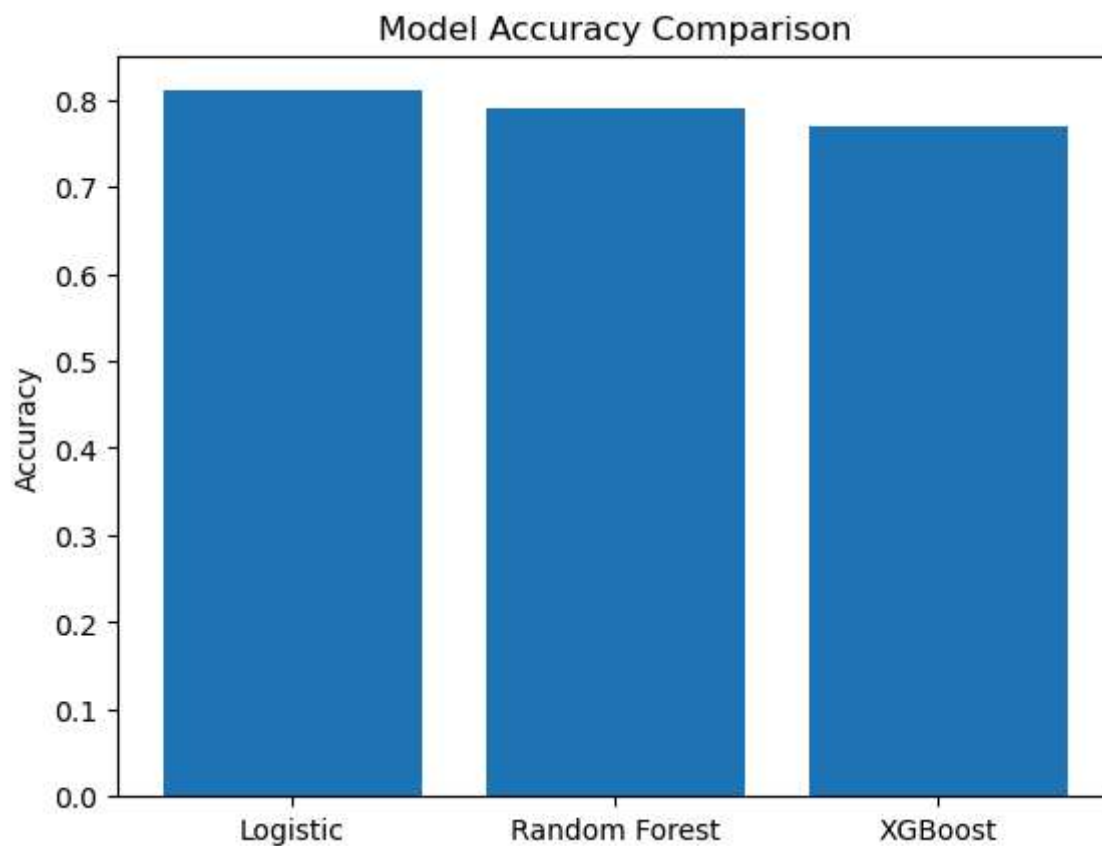
In [99]: # finding all algorithm of accuracy score
print("Logistic:", accuracy_score(y_test, L_pred))
print("Random Forest:", accuracy_score(y_test, R_pred))
print("XGBoost:", accuracy_score(y_test, X_pred))
  
```

Logistic: 0.8097826086956522
 Random Forest: 0.7934782608695652
 XGBoost: 0.7717391304347826

```

In [119]: #graph b/w accuracy score
models = ['Logistic', 'Random Forest', 'XGBoost']
accuracy = [0.81, 0.79, 0.77]

plt.bar(models, accuracy)
plt.title("Model Accuracy Comparison")
plt.ylabel("Accuracy")
plt.show()
  
```



```
In [101... #finnding classificaton_report of all the algorithm
from sklearn.metrics import classification_report
print('Classification Report of Logistic : \n', classification_report(y_test , L_pr
print('Classification Report of Random Forest: \n', classification_report(y_test ,
print('Classification Report of XGBoost : \n', classification_report(y_test , X_pre
```

Classification Report of Logistic :

	precision	recall	f1-score	support
0	0.74	0.83	0.79	77
1	0.87	0.79	0.83	107
accuracy			0.81	184
macro avg	0.81	0.81	0.81	184
weighted avg	0.82	0.81	0.81	184

Classification Report of Random Forest:

	precision	recall	f1-score	support
0	0.75	0.75	0.75	77
1	0.82	0.82	0.82	107
accuracy			0.79	184
macro avg	0.79	0.79	0.79	184
weighted avg	0.79	0.79	0.79	184

Classification Report of XGBoost :

	precision	recall	f1-score	support
0	0.72	0.74	0.73	77
1	0.81	0.79	0.80	107
accuracy			0.77	184
macro avg	0.77	0.77	0.77	184
weighted avg	0.77	0.77	0.77	184

```
In [106... #confusion matrix ao all alogorithum
from sklearn.metrics import confusion_matrix

cmL = confusion_matrix(y_test , L_pred)
cmR = confusion_matrix(y_test , R_pred)
cmX = confusion_matrix(y_test , X_pred)

print('Confusion of Logistic : \n' , cmL)
print('Confusion of Forest : \n' , cmR)
print('Confusion of XGBoost : \n' , cmX)
```

Confusion of Logistic :

```
[[64 13]
 [22 85]]
```

Confusion of Forest :

```
[[58 19]
 [19 88]]
```

Confusion of XGBoost :

```
[[57 20]
 [22 85]]
```

```
In [114... #heat map of all confusion matrix of all alorithums
plt.figure(figsize=(18, 6))

def plotting_heat(cm, title, num):
```

```

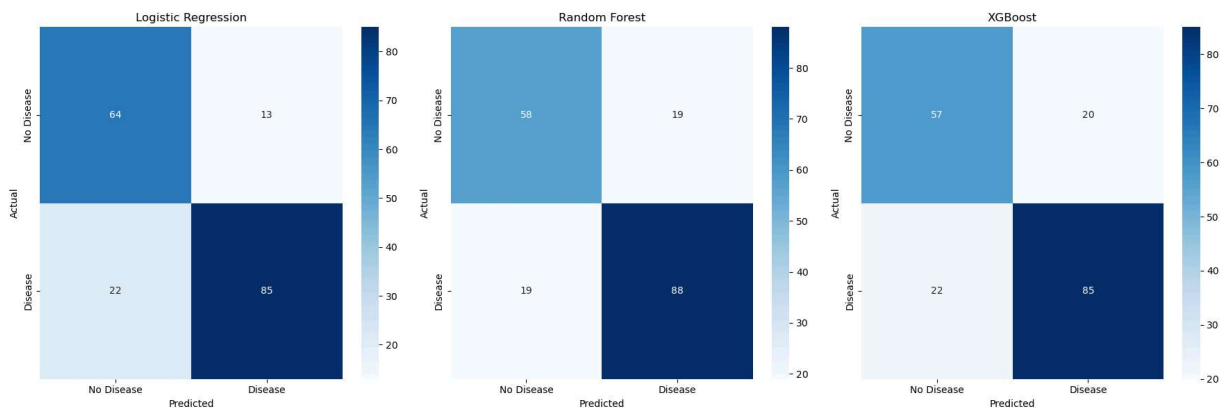
plt.subplot(1, 3, num)
sns.heatmap(cm,
             annot=True,
             fmt='d',
             cmap='Blues',
             xticklabels=['No Disease', 'Disease'],
             yticklabels=['No Disease', 'Disease'])

plt.title(title)
plt.xlabel('Predicted')
plt.ylabel('Actual')

plotting_heat(cml, 'Logistic Regression', 1)
plotting_heat(cmR, 'Random Forest', 2)
plotting_heat(cmX, 'XGBoost', 3)

plt.tight_layout()
plt.show()

```



In [115... `from sklearn.metrics import roc_curve, auc`

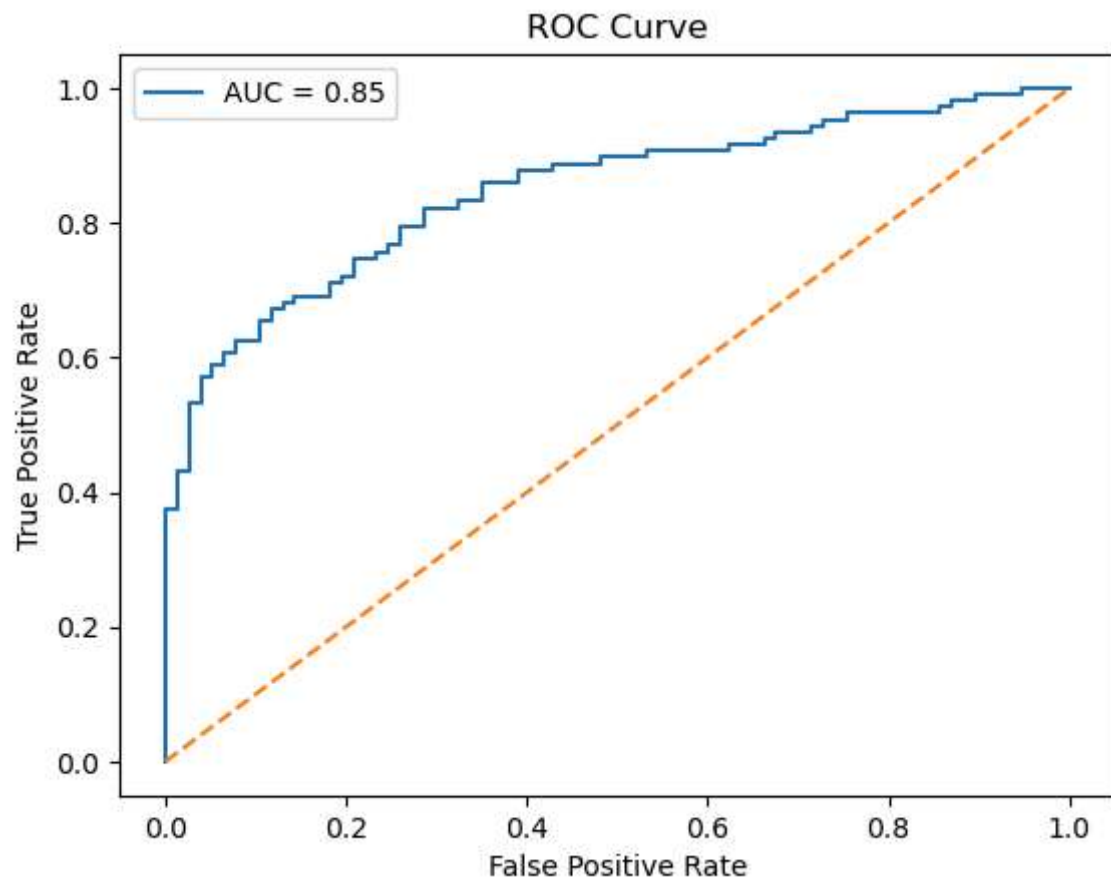
```

y_prob = model.predict_proba(x_test)[: ,1]

fpr, tpr, _ = roc_curve(y_test, y_prob)
roc_auc = auc(fpr, tpr)

plt.plot(fpr, tpr, label=f'AUC = {roc_auc:.2f}')
plt.plot([0,1],[0,1], '--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve')
plt.legend()
plt.show()

```



In []:

In []: